

Lab Meeting

2024/7/24

Hyeongchan Ham

Contents

- Current Works
 - (etc) KAIST Individual study instruct
 - (Research) Out-Of-Distribution Autonomous Driving survey
- Future Plan
 - (Research) Research idea
 - (paper review) TD-MPC2

Current Works

KAIST Individual study instruct

Engineering

Platform		Mac	
Environment		Docker	 Docker
Framework		RLKit	 Safety Gym

Issue

- Architecture mismatch
 - Mac M1 chip AArch64 architecture
- Visualization trouble
 - Use VNC (Virtual Network Computing)
- Library Version compatibility
 - Encapsulate with docker image

Research

benchmark		OpenAI Gym	 Safety Gym
task		Offline RL	Constrained MDP
Current Algorithm		DQN	CPO

Out-Of-Distribution Autonomous Driving

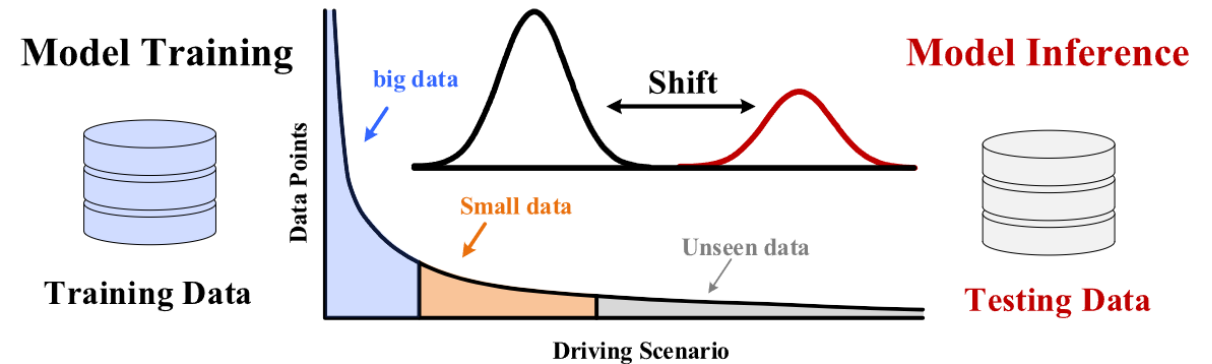
- [367 citations] (arxiv 2017) Kahn, G., Villaflor, A., Pong, V., Abbeel, P., & Levine, S. Uncertainty-aware reinforcement learning for collision avoidance.
- [181 citations] (ICRA 2019) Lütjens, B., Everett, M., & How, J. P. Safe reinforcement learning with model uncertainty estimates
- [93 citations] (RA-L 2019) Hakobyan, A., Kim, G. C., & Yang, I. Risk-aware motion planning and control using CVaR-constrained optimization.
- [24 citations] (ICRA 2020) Hakobyan, A., & Yang, I. Wasserstein distributionally robust motion planning and control with safety constraints using conditional value-at-risk.
- [127 citations] (IEEE IV 2022) Tang, X., Yang, K., Wang, H., Wu, J., Qin, Y., Yu, W., & Cao, D. Prediction-uncertainty-aware decision-making for autonomous vehicles.
- [13 citations] (IEEE ITS 2023) Yang, K., Li, B., Shao, W., Tang, X., Liu, X., & Wang, H. Prediction failure risk-aware decision-making for autonomous vehicles on signalized intersections.

Task Formulation

- AI-driven autonomous driving has high dependency on the training data.
- Long-tailed scenario, like car accident, should be handled safely.



AI-based potential car accident scenario generation [1]



Long-tailed distribution data [2]

[1] <https://developer.nvidia.com/blog/generating-ai-based-accident-scenarios-for-autonomous-vehicles/>

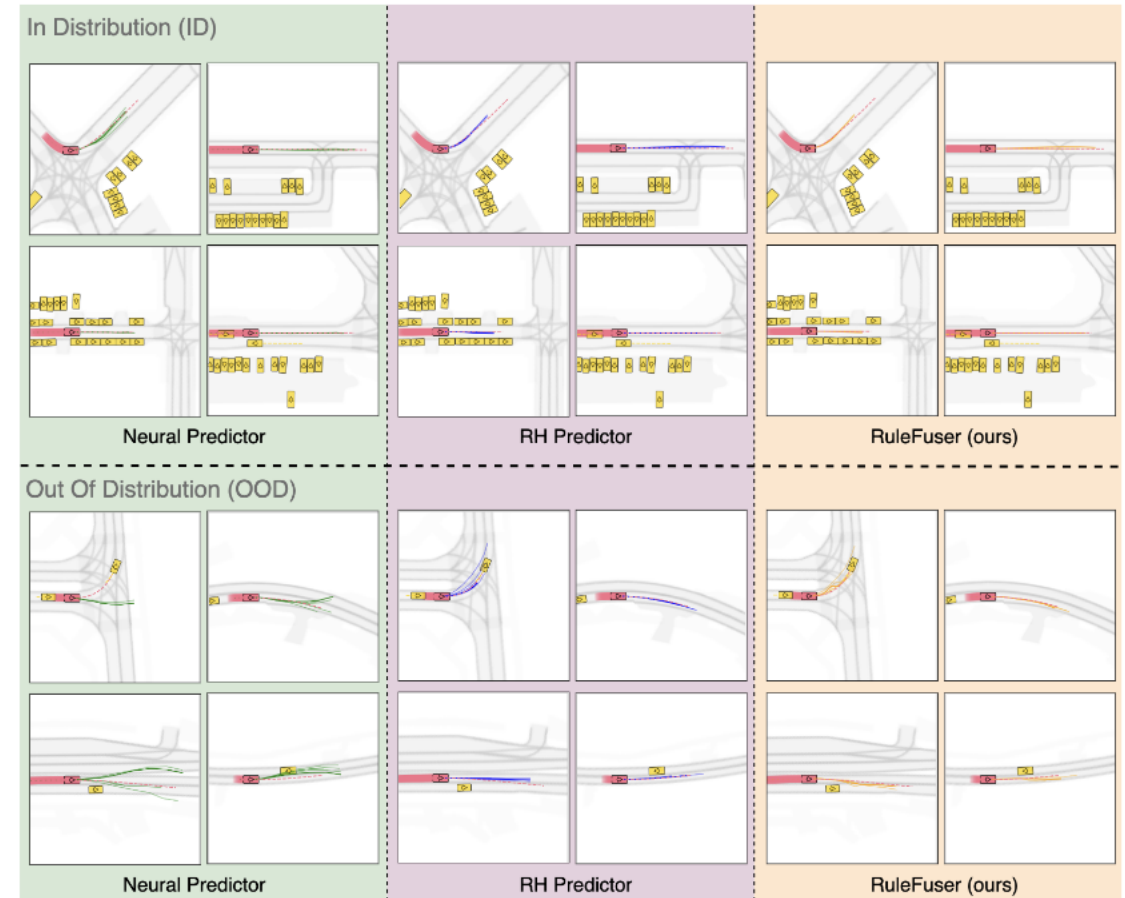
[2] Yang, K., Li, B., Shao, W., Tang, X., Liu, X., & Wang, H. (2023). Prediction failure risk-aware decision-making for autonomous vehicles on signalized intersections. IEEE Transactions on Intelligent Transportation Systems.

Task Formulation

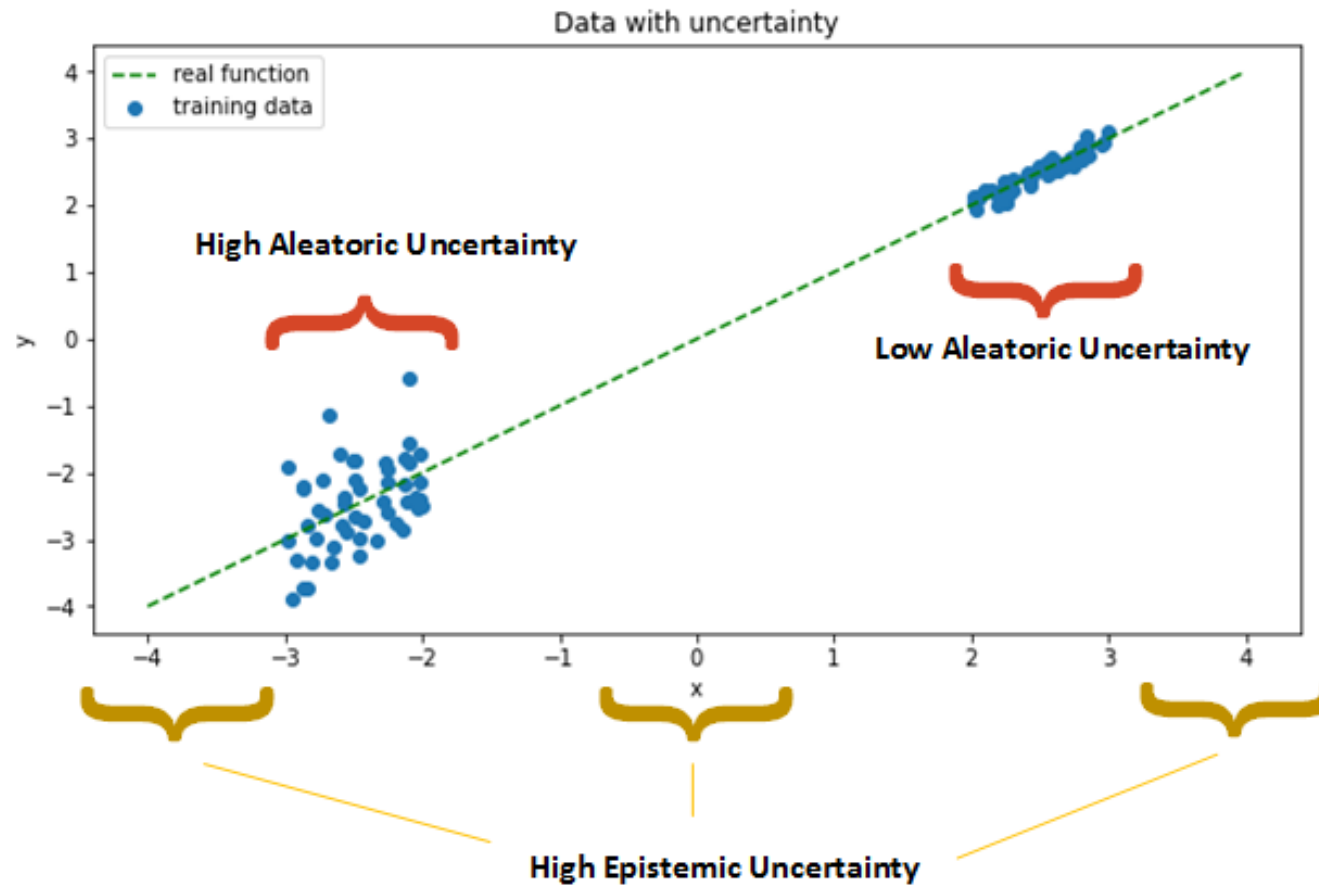
- But, the current trajectory prediction models suffer out-of-distribution (long-tailed) problem.

ID (Boston)
- Drive on the right-side of the road

OOD (Singapore)
- Drive on the left-side of the road



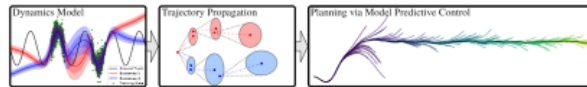
About Uncertainties



Uncertainty application (for exploration)

PETS

- Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models



Algorithm 1 Our model-based MPC algorithm 'PETS':

```

1: Initialize data  $\mathcal{D}$  with a random controller for one trial.
2: for Trial  $k = 1$  to  $K$  do
3:   Train a  $PE$  dynamics model  $\tilde{f}$  given  $\mathcal{D}$ .
4:   for Time  $t = 0$  to TaskHorizon do
5:     for Actions sampled  $\mathbf{a}_{t:t+T} \sim \text{CEM}(\cdot)$ , 1 to NSamples do
6:       Propagate state particles  $\mathbf{s}_t^p$  using  $\tilde{f}$  and  $\mathbf{a}_{t:t+T}$ .
7:       Evaluate actions as  $\sum_{\tau=t}^{t+T} \frac{1}{P} \sum_{p=1}^P r(\mathbf{s}_\tau^p, \mathbf{a}_\tau)$ .
8:       Update  $\text{CEM}(\cdot)$  distribution.
9:     Execute first action  $\mathbf{a}_t^*$  (only) from optimal actions  $\mathbf{a}_{t:t+T}^*$ .
10:    Record outcome:  $\mathcal{D} \leftarrow \mathcal{D} \cup \{\mathbf{s}_t, \mathbf{a}_t^*, \mathbf{s}_{t+1}\}$ .
  
```

- PETS uses ensemble to generate epistemic uncertainty
 - Used for exploration

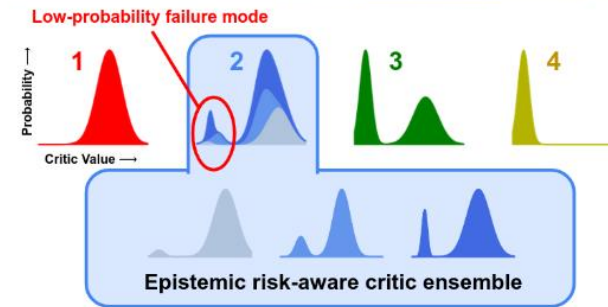


Fig. 4: CVaR naturally accounts for epistemic uncertainty when applied to the mixture distribution output of an ensemble. When the ensemble disagrees about the distribution, the CVaR of their mixture prioritizes more pessimistic ensemble members.

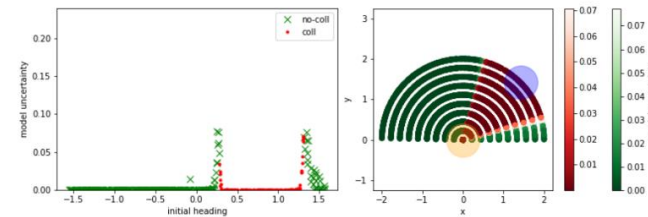


- Risk-sensitive Actor Critic with Epistemic Robustness (RACER) uses ensemble to generate epistemic uncertainty
 - Avoid out-of-distribution states

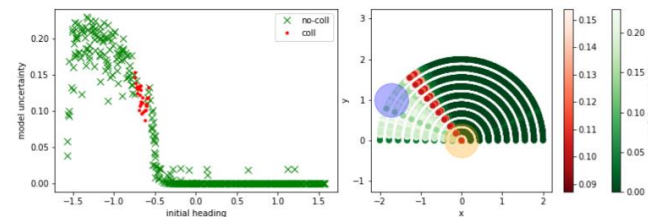
Uncertainty application (for exploration)



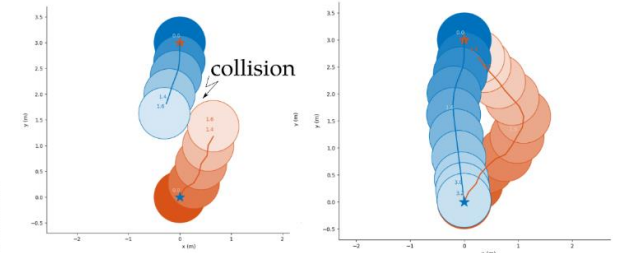
- Uncertainty estimation with bootstrapping and dropout
- Formulates an uncertainty-dependent cost function



(a) Known obstacle: low uncertainty



(b) Novel obstacle: high uncertainty



(a) uncertainty-unaware

(b) uncertainty-aware

- Uncertainty estimation with MC-Dropout and Bootstrapping
- Policy explores in directions of high epistemic uncertainty during early training phases

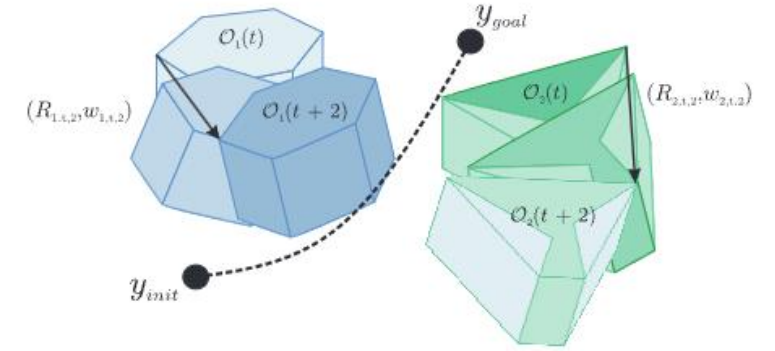
[6] Kahn, G., Villafior, A., Pong, V., Abbeel, P., & Levine, S. (2017). Uncertainty-aware reinforcement learning for collision avoidance. arXiv preprint arXiv:1702.01182.

[7] Lütjens, B., Everett, M., & How, J. P. (2019, May). Safe reinforcement learning with model uncertainty estimates. In 2019 International Conference on Robotics and Automation (ICRA) (pp. 8662-8668). IEEE.

- However, the lack of interpretability of learning-based decision-making methods will cause safety concerns.
- In contrast, model-based approaches can achieve a degree of transparency in a planner's decision-making.

Hakobyan et al.

- Dynamic obstacles avoidance task
- CVaR based motion planning and control
- Computationally feasible
 - CVaR constraint reformulation
 - Sample Average Approximation
 - Linearly Constrained Mixed Integer Convex Program



obstacle $\mathcal{O}_\ell^o(t)$

robot's position $y(t)$

safe region $\mathcal{Y}_\ell(t) = \mathbb{R}^{n_y} \setminus \mathcal{O}_\ell^o(t)$

$$\text{dist}(y(t), \mathcal{Y}_\ell(t)) := \min \|y(t) - a\|$$

$$\min_{\mathbf{u}, \mathbf{x}, \mathbf{y}} J(x(t), \mathbf{u}) := \sum_{k=0}^{K-1} r(x_k, u_k) \quad (5a)$$

$$\text{s.t. } x_{k+1} = Ax_k + Bu_k \quad (5b)$$

$$y_k = Cx_k + Du_k \quad (5c)$$

$$x_0 = x(t) \quad (5d)$$

$$x_k \in \mathcal{X}, u_k \in \mathcal{U} \quad (5e)$$

$$\text{CVaR}_\alpha[\text{dist}(y_k, \mathcal{Y}_\ell(t+k))] \leq \delta_\ell \quad \forall \ell, \quad (5f)$$

[9] Hakobyan, A., Kim, G. C., & Yang, I. (2019). Risk-aware motion planning and control using CVaR-constrained optimization. IEEE Robotics and Automation letters, 4(4), 3924-3931.

[10] Hakobyan, A., & Yang, I. (2020, May). Wasserstein distributionally robust motion planning and control with safety constraints using conditional value-at-risk.

In 2020 IEEE International Conference on Robotics and Automation (ICRA) (pp. 490-496). IEEE.

Tang et al.

- Used LSTM based ensemble to generate epistemic / aleatoric uncertainty
- Uncertainty-aware potential field (UPF) based on the uncertainty (epistemic + aleatoric)
- With this UPF, MPC-based decision making framework

Road Potential Field

$$PF = \sum UPF_i + \sum RPF_k$$

$$UPF_i(X, Y) = a_i e^{\beta_i}$$

$$\beta_i = - \left\{ \frac{((X - E_{x*})\cos\vartheta + (Y - E_{y*})\sin\vartheta)^2}{2\lambda_{x_i}^2} + \frac{(-(X - E_{x*})\sin\vartheta + (Y - E_{y*})\cos\vartheta)^2}{2\lambda_{y_i}^2} \right\}^{b_i}$$

$$\lambda_{x_i} = L_{x_i} + M^{-1} \sum_{m=1}^M (\sigma_x^2 + E_x^2)_m - E_{x*}^2$$

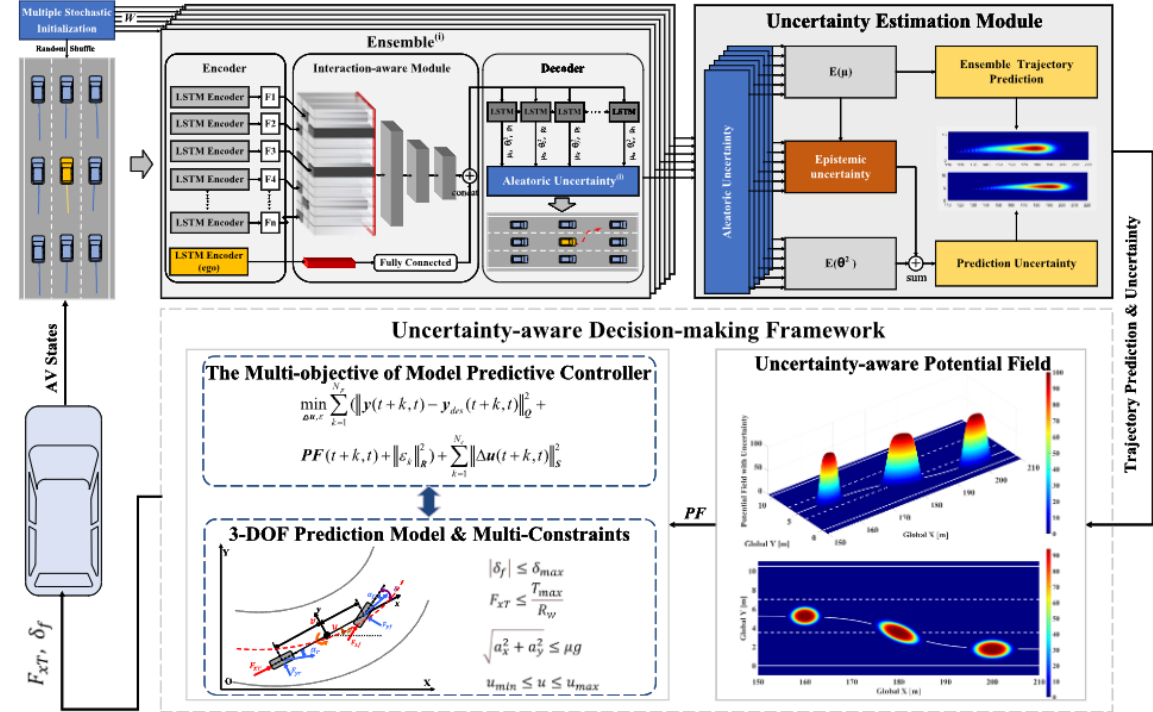
$$\lambda_{y_i} = L_{y_i} + M^{-1} \sum_{m=1}^M (\sigma_y^2 + E_y^2)_m - E_{y*}^2$$

Each network m in the ensemble model

$$\mathcal{R}(\mathcal{X}) = [E_m, cov_m]$$

$$E_m = [E_x, E_y]_m$$

$$cov_m = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}_m$$

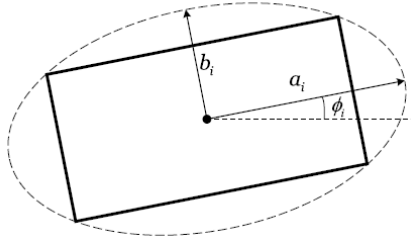


Ensemble output

$$\begin{aligned} E_{y*}(\mathcal{X}) &= [E_{x*}, E_{y*}] = M^{-1} \sum_{m=1}^M E_m \\ cov_{y*}(\mathcal{X}) &= E_{y*}^T cov_{y*}(\mathcal{X}) - E_{y*}(\mathcal{X})^T E_{y*}(\mathcal{X}) \\ &= M^{-1} \sum_{m=1}^M \begin{bmatrix} \sigma_x^2 + E_x^2 & \rho\sigma_x\sigma_y + E_x E_y \\ \rho\sigma_x\sigma_y + E_x E_y & \sigma_y^2 + E_y^2 \end{bmatrix}_m \\ &\quad - \begin{bmatrix} E_{x*}^2 & E_{x*} E_{y*} \\ E_{x*} E_{y*} & E_{y*}^2 \end{bmatrix}_m \end{aligned} \quad (18)$$

RADM (Risk-aware decision-making)

- Multi-agent prediction network with ensemble generates epistemic uncertainty
- Based on quantified epistemic uncertainty, the prediction failure risk is calculated
- MPC-based decision-making incorporates both prediction results and failure risk



$$obs_cons_t^{i,cj}(x_t^E, x_t^{O_i}) = \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix} > 1$$

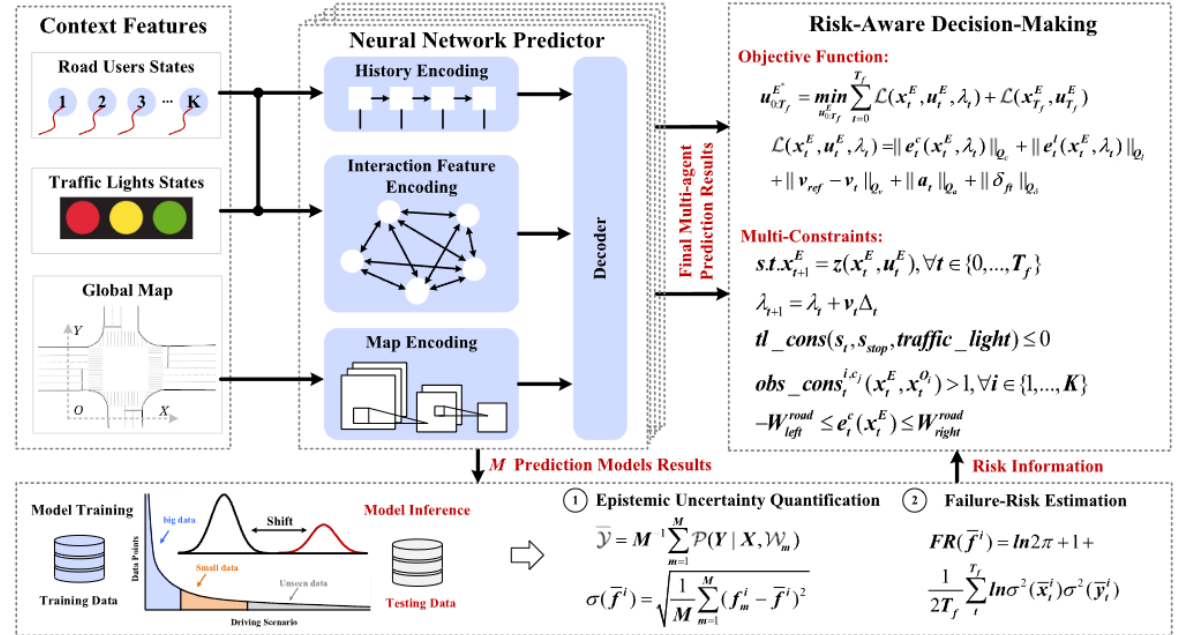
Enlarge OV's semi-major and semi-minor axes with a fixed margin

$$obs_cons_t^{i,cj}(x_t^E, x_t^{O_i}) = \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix} > 1$$

$$\bar{a}_t^i = a^i + \eta \sigma(\bar{x}_t^i), \quad \bar{b}_t^i = b^i + \mu \sigma(\bar{y}_t^i)$$

Too conservative! (especially for dense traffic scenarios, such as intersections)

Self-adaptively adjust the model parameters instead of fully trusting the prediction results



RADM (Risk-aware decision-making)

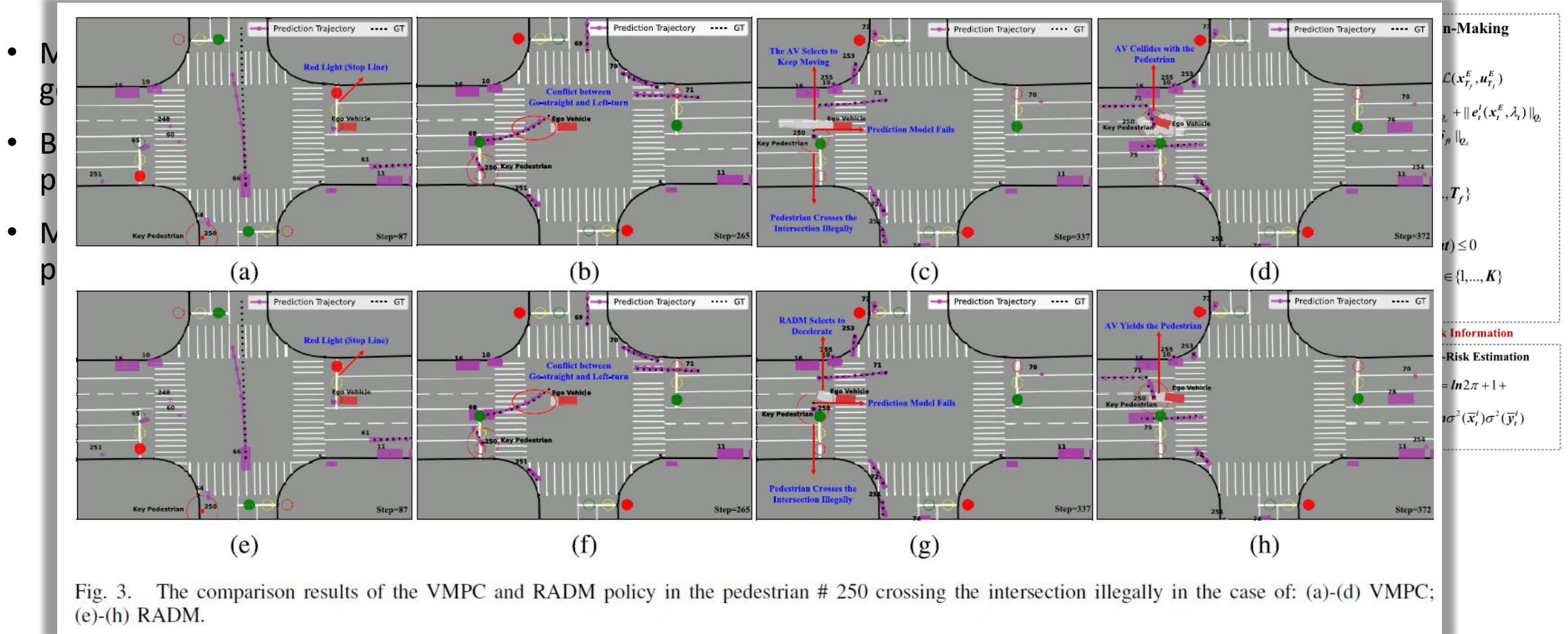


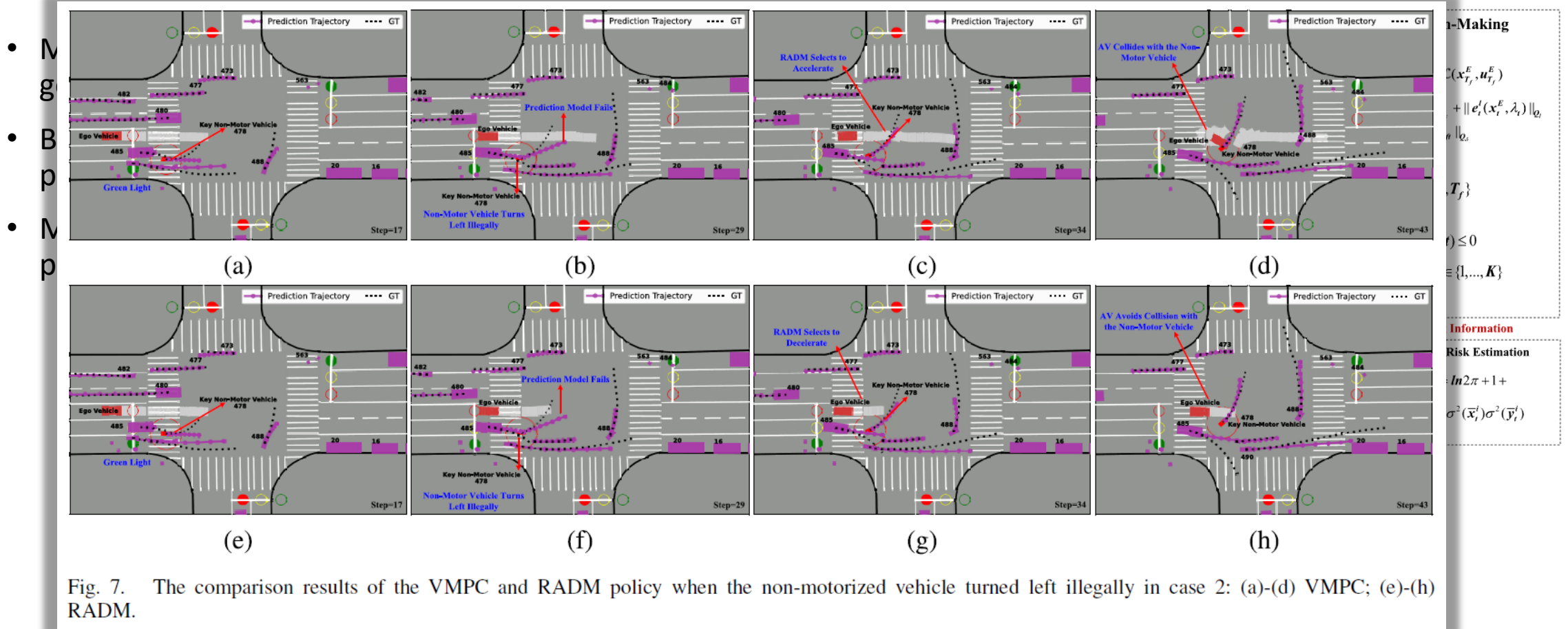
Fig. 3. The comparison results of the VMPC and RADM policy in the pedestrian # 250 crossing the intersection illegally in the case of: (a)-(d) VMPC; (e)-(h) RADM.

$$= \begin{bmatrix} \Delta x_{t_i}^{i,cj} \\ \Delta y_{t_i}^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_{t_i}^{i,cj} \\ \Delta y_{t_i}^{i,cj} \end{bmatrix} > 1$$

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Self-adaptively adjust the model parameters instead of fully trusting the prediction results

RADM (Risk-aware decision-making)



$$= \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix} > 1$$

$$\bar{a}_t^i = a^i + \eta \sigma(\bar{x}_t^i), \quad \bar{b}_t^i = b^i + \mu \sigma(\bar{y}_t^i)$$

Self-adaptively adjust the model parameters instead of fully trusting the prediction results

RADM (Risk-aware decision-making)

- N
- B
- N

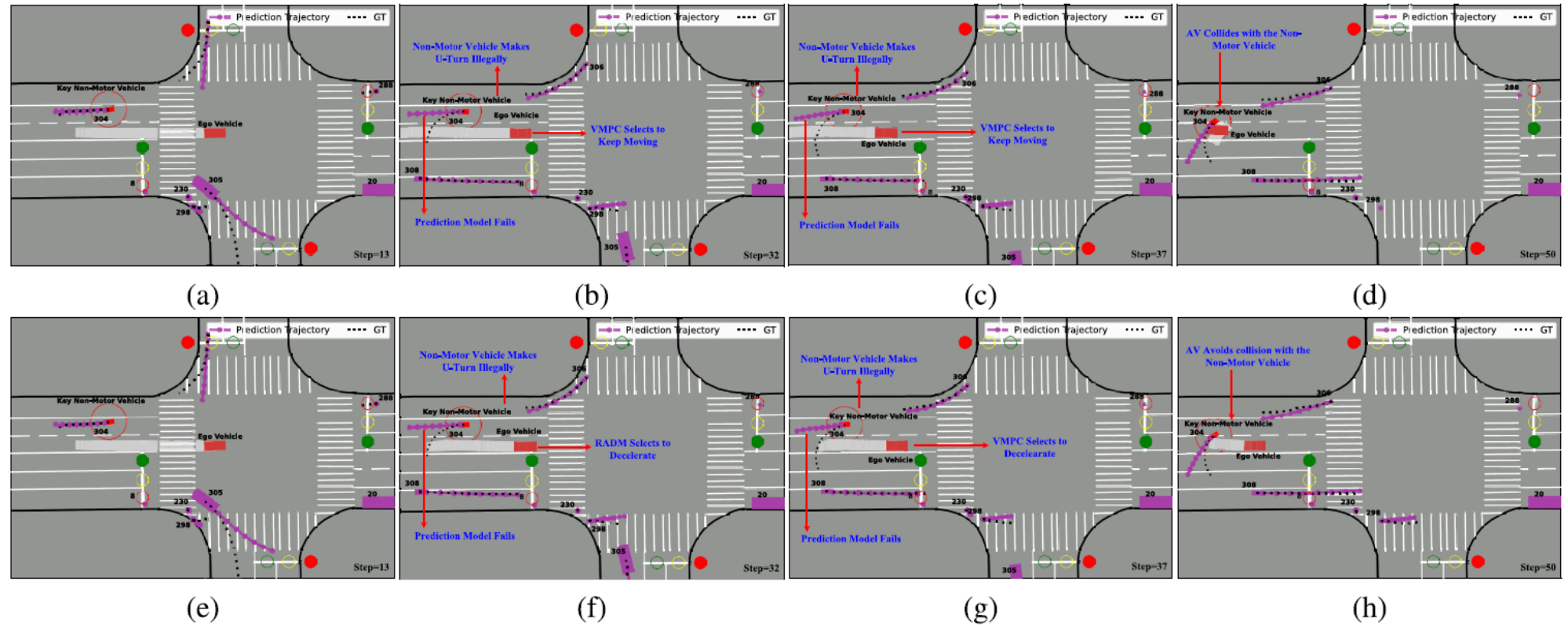


Fig. 11. The comparison results of the VMPC and RADM policy of a non-motorized vehicle making the U-Turn illegally in case 3: (a)-(d) VMPC; (e)-(h) RADM.

$$= \begin{bmatrix} \Delta x_{t,i,cj}^{i,cj} \\ \Delta y_{t,i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^2} & 0 \\ 0 & \frac{1}{\bar{b}_t^2} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_{t,i,cj}^{i,cj} \\ \Delta y_{t,i,cj} \end{bmatrix} > 1$$

$$\bar{a}_t^i = a^i + \eta \sigma(\tilde{x}_t^i), \quad \bar{b}_t^i = b^i + \mu \sigma(\tilde{y}_t^i)$$

Self-adaptively adjust the model parameters instead of fully trusting the prediction results

-Making

$$(x_{t_f}^E, u_{t_f}^E) + \|e_t'(x_t^E, \lambda_t)\|_{Q_t}$$

$T_f\}$

≤ 0

$\{1, \dots, K\}$

Information

Risk Estimation

$$\ln 2\pi + 1 +$$

$$\tau^2(\bar{x}_t^i)\sigma^2(\bar{y}_t^i)$$

RADM (Risk-aware decision-making)

- Multi-agent prediction network with ensemble generates epistemic uncertainty

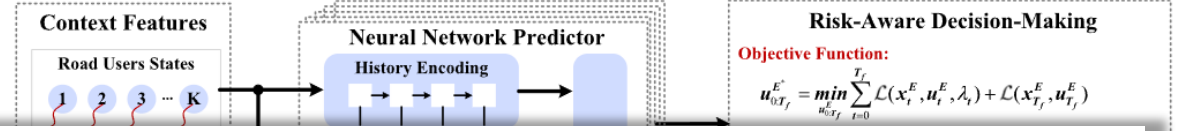


TABLE III

QUANTITATIVE PERFORMANCE EVALUATION RESULTS OBTAINED BY THE SUMO SIMULATOR

Scenario	Metrics	VMPC	CMPC	RADM
Normal scenario	Collision Rate (%)	3.80 ± 2.04	2.80 ± 0.75	3.00 ± 1.10
	Traffic Violation Rate (%)	0	0	0
	Average Speed (m/s)	7.40 ± 0.06	5.47 ± 0.07	7.10 ± 0.09
Challenging scenario	Collision Rate (%)	14.20 ± 2.48	7.40 ± 2.80	7.20 ± 2.48
	Traffic Violation Rate (%)	0	0	0
	Average Speed (m/s)	6.97 ± 0.24	4.12 ± 0.44	6.25 ± 0.01

*These methods are tested for 5 epochs with 5 different random seeds and in each epoch, they are tested 100 times.

$$obs_cons_t^{i,cj}(x_t^E, x_t^{O_i})$$

$$= \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix} > 1$$

Enlarge OVS semi-major and semi-minor axes with a fixed margin

Too conservative! (especially for dense traffic scenarios, such as intersections)

$$obs_cons_t^{i,cj}(x_t^E, x_t^{O_i})$$

$$= \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix}^T R(\phi_i)^T \begin{bmatrix} \frac{1}{\bar{a}_t^{i2}} & 0 \\ 0 & \frac{1}{\bar{b}_t^{i2}} \end{bmatrix} R(\phi_i) \begin{bmatrix} \Delta x_t^{i,cj} \\ \Delta y_t^{i,cj} \end{bmatrix} > 1$$

$$\bar{a}_t^i = a^i + \eta\sigma(\tilde{x}_t^i), \quad \bar{b}_t^i = b^i + \mu\sigma(\tilde{y}_t^i)$$

Self-adaptively adjust the model parameters instead of fully trusting the prediction results

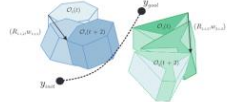
Future Plan

Out-Of-Distribution Autonomous Driving

- Guarantee with the perception uncertainty estimation?

Hakobyan et al.

- Dynamic obstacles avoidance task
- CVaR based motion planning and control
- Computationally feasible
 - CVaR constraint reformulation
 - Sample Average Approximation
 - Linearly Constrained Mixed Integer Convex Program



obstacle $O_i(t)$
robot's position $y(t)$
safe region $\mathcal{Y}_i(t) = \mathbb{R}^{n_x} \setminus O_i(t)$

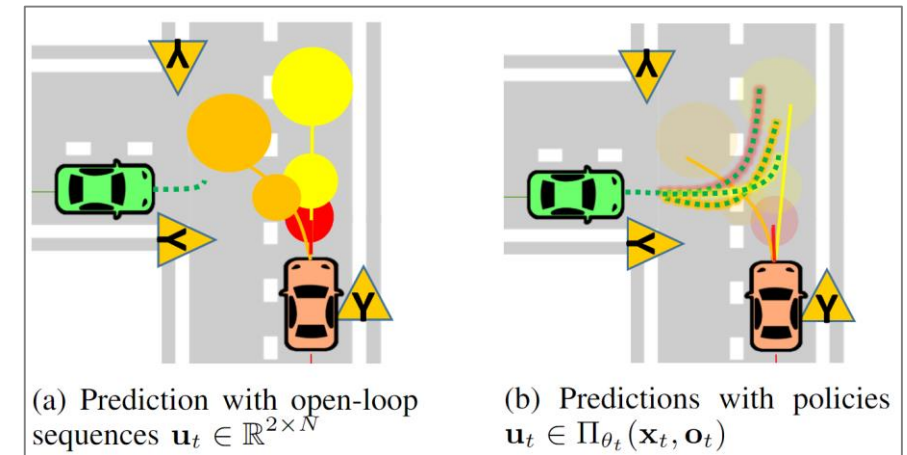
$$\text{dist}(y(t), \mathcal{Y}_i(t)) := \min \|y(t) - a\|$$

$$\min_{u, x, y} J(x(t), u) := \sum_{k=0}^{K-1} r(x_k, u_k) \quad (5a)$$

s.t. $x_{k+1} = Ax_k + Bu_k \quad (5b)$
 $y_k = Cx_k + Du_k \quad (5c)$
 $x_0 = x(t) \quad (5d)$
 $x_k \in \mathcal{X}, u_k \in \mathcal{U} \quad (5e)$
 $\text{CVaR}_\alpha[\text{dist}(y_k, \mathcal{Y}_i(t+k))] \leq \delta_\ell \quad \forall \ell \quad (5f)$

[1] Hakobyan, A., Kim, G. C., & Yang, I. (2019). Risk-aware motion planning and control using CVaR-constrained optimization. IEEE Robotics and Automation letters, 4(4), 3924-3931.
 [2] Hakobyan, A., & Yang, I. (2020, May). Wasserstein distributionally robust motion planning and control with safety constraints using conditional value-at-risk. In 2020 IEEE International Conference on Robotics and Automation (ICRA) (pp. 490-496). IEEE.

Research Position



Zero-knowledge

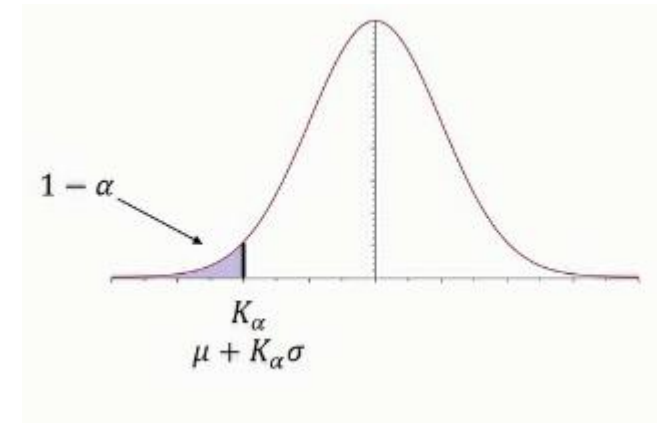
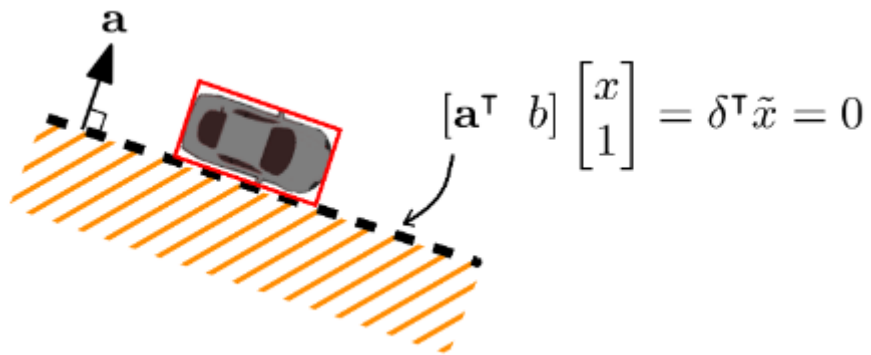
Full-knowledge

[9] Hakobyan, A., Kim, G. C., & Yang, I. (2019). Risk-aware motion planning and control using CVaR-constrained optimization. IEEE Robotics and Automation letters, 4(4), 3924-3931.

[10] Nair, S. H., Govindarajan, V., Lin, T., Meissen, C., Tseng, H. E., & Borrelli, F. (2022, October). Stochastic mpc with multi-modal predictions for traffic intersections.

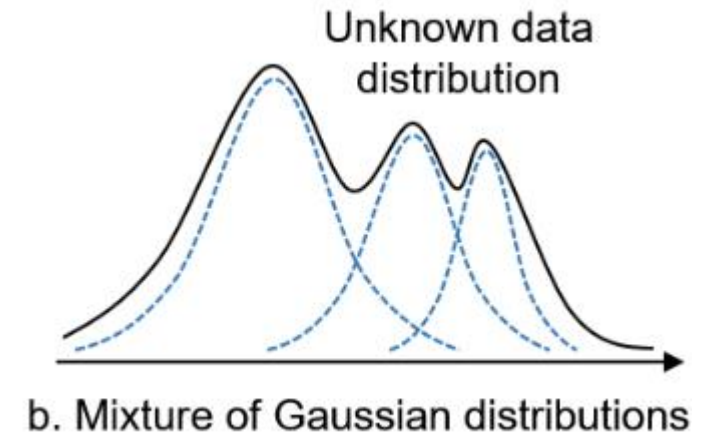
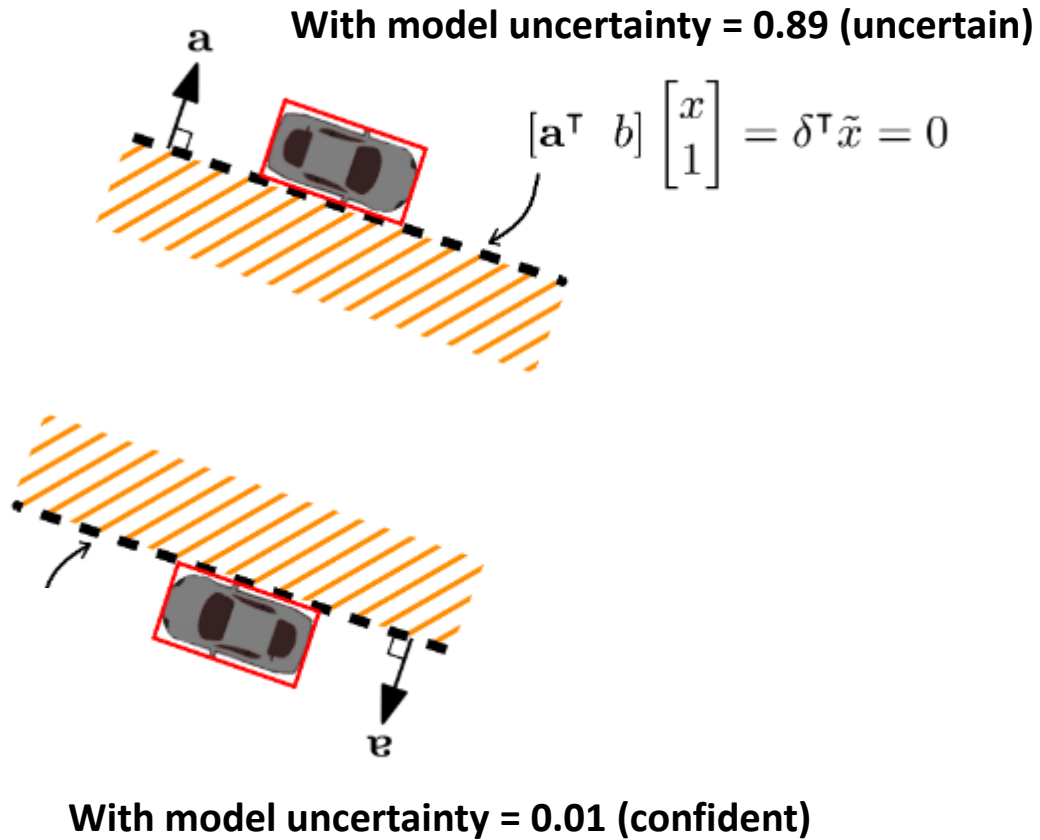
In 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC) (pp. 635-640). IEEE.

Out-Of-Distribution Autonomous Driving

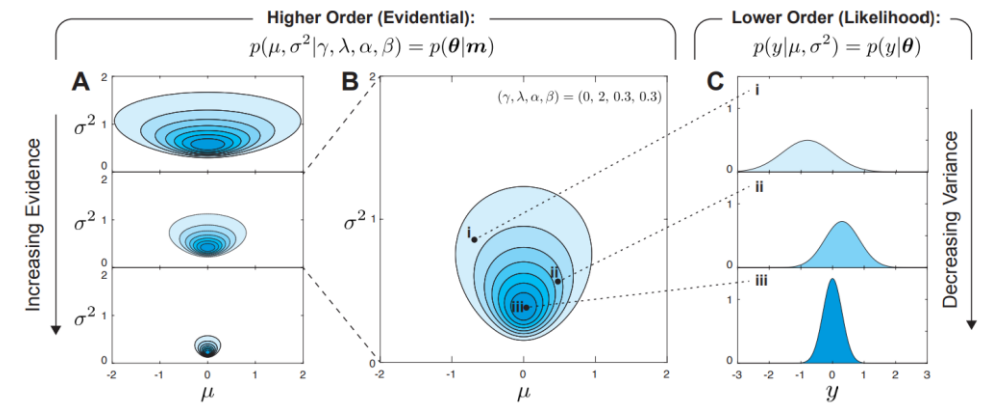
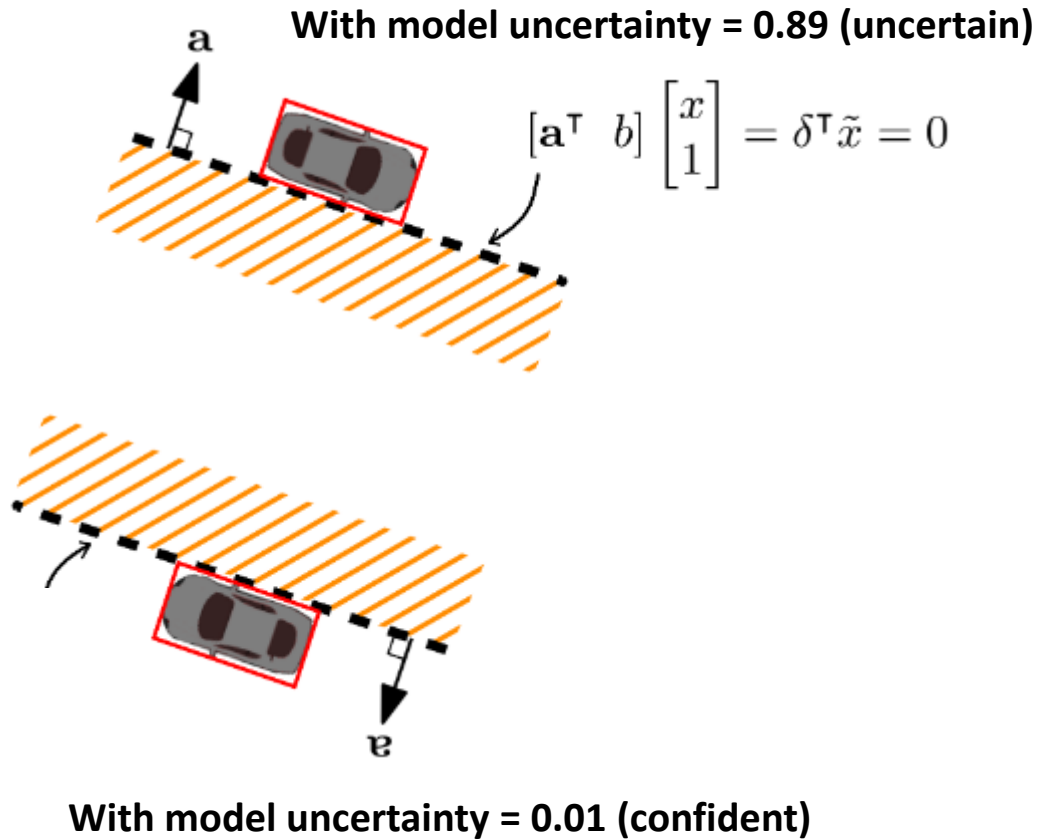


Fully trust perception result (no model uncertainty, only data uncertainty)

Out-Of-Distribution Autonomous Driving

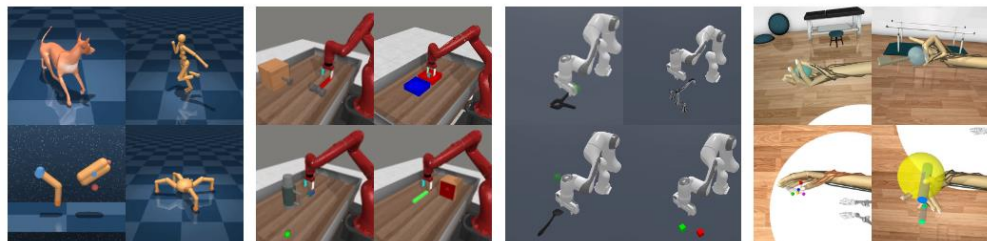


Out-Of-Distribution Autonomous Driving

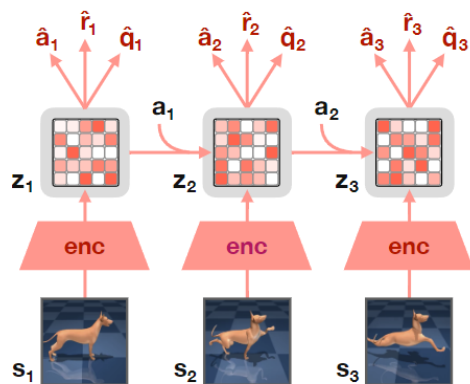


Guarantee with higher order distribution?
 - Parameter of parameters

TD-MPC2



Various tasks (104 tasks)



TD-MPC2 architecture

